

Cool Things You Need to Know About Triangles

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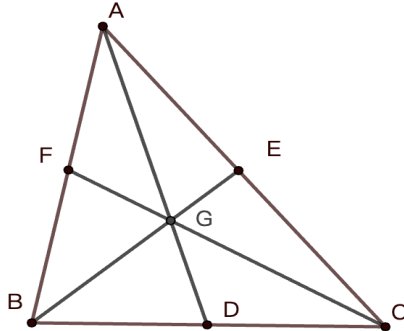
1 Why are triangles so important?

In geometry, polygons are everywhere. For instance, even though a circle is not a defined polygon, it can still be considered as a polygon that has infinite sides. Since any polygon can be formed by a number of triangles, the triangle becomes one of the most important shapes in geometry. (Think like this: all polygons can be cut to a bunch of triangles.) This article will introduce a series of triangle properties, including the centroid, incenter, orthocenter, and some famous theorems related to triangles, such as the Angle Bisector Theorem, Law of Cosines, and Stewart's Theorem.

2 Cevians

In order to talk about the properties of triangles and their applications, we have to know what a cevian is: A cevian is a line segment or ray that extends from one vertex of a triangle to the opposite side. Angle bisectors, altitudes, and medians of a triangle are all special types of cevians. One famous theorem related to cevians is Ceva's theorem, which states that in any $\triangle ABC$:

$$\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1$$



2.1 Proof of Ceva's Theorem

Note: $[ABC]$ indicates the area of triangle ABC. Notice that when triangles share the same height, the ratio of their areas is simply the ratio of the lengths of their bases.

$$\frac{AF}{FB} = \frac{[AGF]}{[BGF]}$$

$$\frac{AF}{FB} = \frac{[ACF]}{[BCF]}$$

A quick proof of the baseball theorem in case you are wondering what that is:

Statement: If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a}{b} = \frac{c}{d} = \frac{a-c}{b-d}$.

Proof: $\frac{a}{b} = \frac{c}{d} \rightarrow ad = bc$. Subtract cd from both sides: $ad - cd = bc - cd \rightarrow (a-c)d = (b-d)c \rightarrow \frac{a-c}{d-d} = \frac{c}{d}$, which also equals to $\frac{a}{b}$.

Applying baseball theorem, we get:

$$\frac{[ACF] - [AGF]}{[BCF] - [BGF]} = \frac{AF}{FB}$$

$$\frac{[ACG]}{[CBG]} = \frac{AF}{FB}$$

Applying the same method on $\frac{BD}{DC}$ and $\frac{CE}{EA}$, we obtain:

$$\frac{BD}{DC} = \frac{[AGB]}{[AGC]}, \frac{CE}{EA} = \frac{[BGC]}{[GBA]}$$

Multiply all three area ratios and obtain:

$$\frac{[ACG]}{[CBG]} \cdot \frac{[AGB]}{[AGC]} \cdot \frac{[BGC]}{[GBA]} = 1$$

3 Incenter, Centroid, and Orthocenter

Before talking about these terms, let's first start with the following definitions:

An **angle bisector** is a cevian which divides the given vertex's angle into two congruent angles.

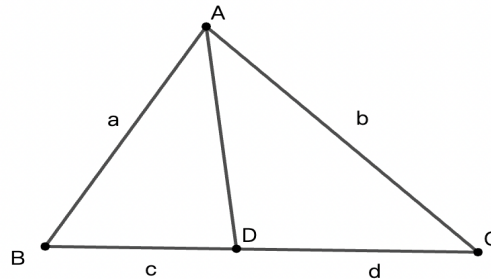
A **median** is a cevian which intersects at the midpoint of its opposite side.

An **altitude** is a cevian which is perpendicular to its opposite side.

3.1 Angle Bisector Theorem

The Angle Bisector theorem is commonly used in triangles, which can be used to find the relationships between segments of the triangle, given there is an angle bisector.

This theorem states that for any triangle ABC, angle bisector AD divides the opposite side in a ratio of AB:AC.

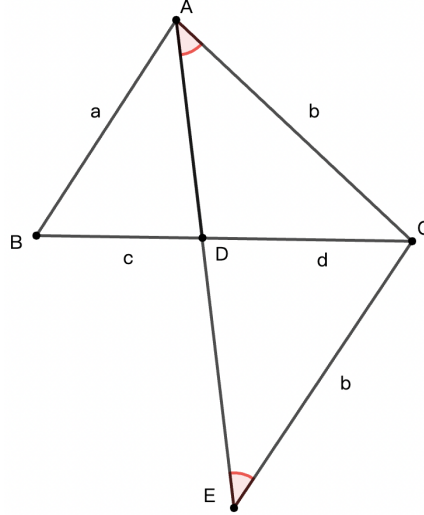


Written as an equation:

$$\frac{AB}{BC} = \frac{AC}{DC} \longrightarrow \frac{a}{c} = \frac{b}{d}$$

3.2 Proof of the Angle Bisector Theorem

There are many different ways to prove Angle Bisector theorem. Here is one simple method: Extend angle bisector AD until a point E ($E \neq A$) such that $CA = CE$.



Since $CA = CE$, $\triangle ACE$ is an isosceles triangle, meaning $\angle CAE \cong \angle CEA$. Since $\angle CAE \cong \angle DAB$, by the definition of an angle bisector, $\angle CEA \cong \angle DAB$, which proves $AB \parallel CE$. Because $AB \parallel CE$, $\angle ABD \cong \angle DCE$. Using AA congruence with $\angle ABD \cong \angle DCE$ and $\angle CAE \cong \angle DAB$, $\triangle ADB \sim \triangle EDC$. Using $\triangle ADB \sim \triangle EDC$:

$$\frac{c}{a} = \frac{d}{b} \longrightarrow \frac{a}{c} = \frac{b}{d}$$

This is equivalent to the angle bisector theorem statement at section 3.1: $\frac{a}{c} = \frac{b}{d}$.

3.3 Brief Introduction

An **incenter** is the point of intersection of three angle bisectors of a triangle, which is also the center of the triangle's incircle.

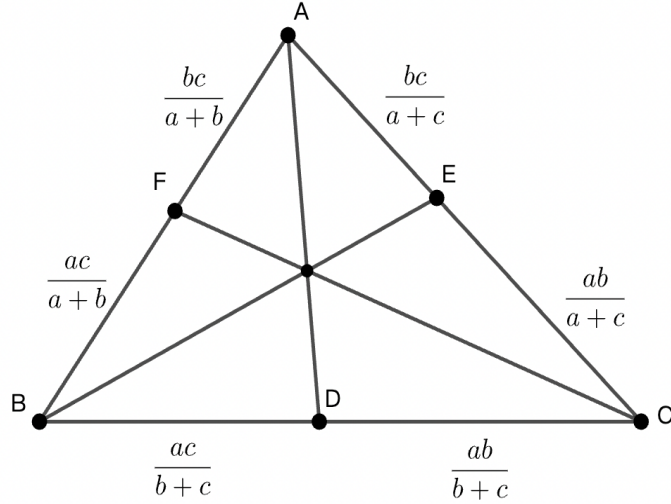
A **centroid** is the point of intersection of three medians of a triangle, which divides each median to a 1 : 2 ratio.

An **orthocenter** is the point of intersections of three altitudes of a triangle.

4 Incenter

I believe the incenter is more interesting to talk about, so I decided to write more about it in this article.

You may wonder why three angle bisectors intersect at the same point, here you go:



Here is the setup:

Let $BC = a$, $AC = b$, $AB = c$.

Let $\overline{AD}$ and $\overline{CF}$ to be angle bisectors. Since we do not know if $\overline{BE}$ is an angle bisector, we need to treat it as a cevian that pass through the intersection point of $\overline{AD}$ and $\overline{CF}$.

Apply angle bisector theorem on angle bisector $\overline{AD}$ and $\overline{CF}$, we can obtain:

$$BD : DC = \frac{ac}{b+c} : \frac{ab}{b+c} \text{ and } AF : FB = \frac{bc}{a+b} : \frac{ac}{a+b}.$$

Assume $\overline{BE}$ is an angle bisector, then angle bisector theorem will hold true, otherwise it will not.

If $\overline{BE}$ is an angle bisector:

$$AE : EC = \frac{bc}{a+c} : \frac{ab}{a+c}.$$

If this is valid, then Ceva's theorem needs to be hold true also:

$$\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} \stackrel{?}{=} 1$$

Plug in those side lengths obtained from applying angle bisector theorem:

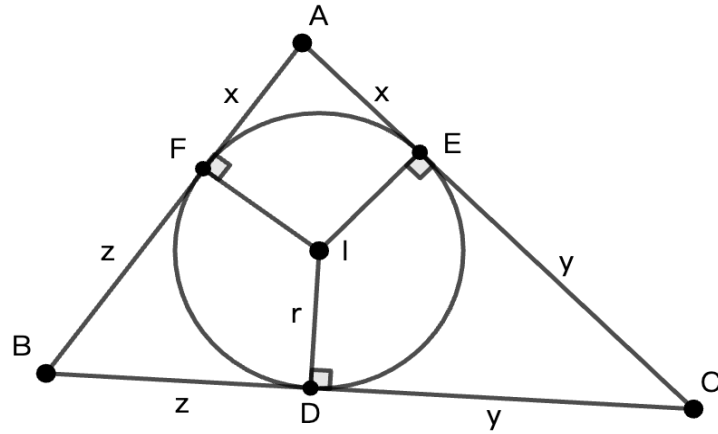
$$\begin{aligned} \frac{\frac{bc}{a+b}}{\frac{ac}{a+b}} \cdot \frac{\frac{ac}{b+c}}{\frac{ab}{b+c}} \cdot \frac{\frac{ab}{a+c}}{\frac{bc}{a+c}} &\stackrel{?}{=} 1 \\ \frac{bc}{ac} \cdot \frac{ac}{ab} \cdot \frac{ab}{bc} &\stackrel{?}{=} 1 \\ 1 &= 1 \end{aligned}$$

Therefore, if $\overline{BE}$ is not an angle bisector, then the ratio of $AE : EC$ is not $\frac{bc}{a+c} : \frac{ab}{a+c}$, which will make Ceva's theorem not holds true, causing a contradiction.

Conclusion: Three angle bisectors of a triangle are **concurrent**.

4.1 Interesting property of incircle

As mentioned in section 3.3, the incenter of a triangle is also the center of its incircle. The incircle of a triangle is the circle that is inscribed inside of a triangle. Here how an incircle looks like:



The incenter is indicated by the letter I . Points D, E, F are the feet of the segment dropped from I . r , the inradius of the triangle, which is the radius of the incircle, equals to the measure of any of ID, IE , and IF .

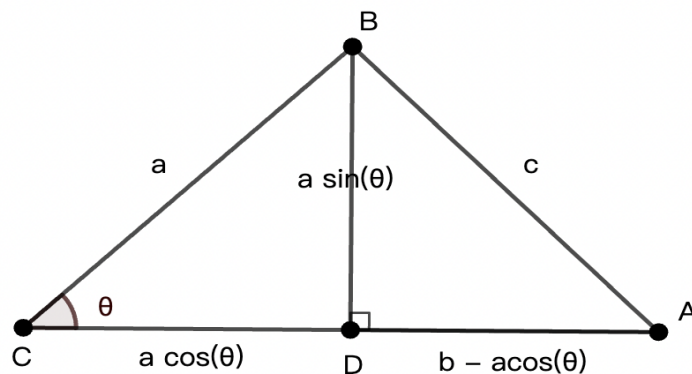
Points D, E, F divide each side by a certain ratio. As indicated in the diagram, $AF = AE$, $CE = CD$, $BD = BF$.

5 Law of Cosines

Law of cosines is an useful theorem which creates a relationship between three sides and one of three angles in a triangle. Using law of cosine, we are able to figure out any side length given one angle and two of the side lengths, or any angle measurement given three side lengths.

5.1 Proof for acute triangle

Consider a triangle ABC , and call angle ACB θ .



First, drop an altitude from B to $\overline{CA}$, call the foot D.

Since ABD is a right triangle, the length of $\overline{BD}$ is $a \sin(\theta)$ and the length of $\overline{CD}$ is $a \cos(\theta)$, which means the length of $\overline{DA}$ is $b - a \cos(\theta)$.

$\triangle BDA$ is also a right triangle, meaning we can apply Pythagorean theorem on it to obtain the following relationship: $(b - a \cos(\theta))^2 + (a \sin(\theta))^2 = c^2$.

Expand this, we have:

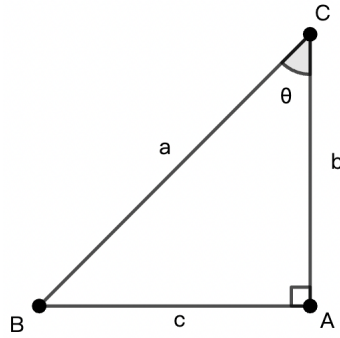
$$b^2 - 2ab \cos(\theta) + a^2 \cos^2(\theta) + a^2 \sin^2(\theta) = c^2$$

$$b^2 - 2ab \cos(\theta) + a^2 = c^2$$

Here we have the statement for law of cosine: $c^2 = a^2 + b^2 - 2ab \cos(\theta)$.

5.2 Proof for right triangle

Since law of cosine can be applied to all triangles and I have only shown the case for acute triangle, we need to know if it works for the case of right triangle.



Starting from a similar setup, law of cosine obviously holds true when θ is the right angle. Therefore, we only need to consider when θ is one of the other two angles.

Knowing that $b = a \cos(\theta)$ and $c = a \sin(\theta)$, we can derive from $2b^2 = 2b^2$:

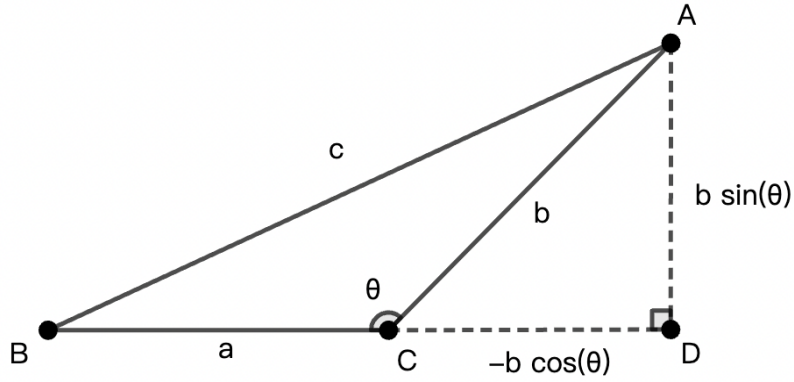
$$c^2 + 2b^2 = b^2 + a^2$$

$$c^2 = a^2 + b^2 - 2b^2$$

$$c^2 = a^2 + b^2 - 2ab \cos(\theta)$$

5.3 Proof for obtuse triangle

Also using the similar setup, we are going to prove if law of cosine holds for the obtuse angle:



First, extended side $\overline{BC}$ to point D and drop an altitude from point A to point D.

Knowing that $\cos(\pi - \theta) = -\cos(\theta)$, we know that the length of $\overline{CD}$ is $-b \cos(\theta)$. Also since $\sin(\theta) = \sin(\pi - \theta)$, the length of $\overline{AD}$ is $b \sin(\theta)$.

Applying Pythagorean theorem on triangle ABD, we have: $(a - b \cos(\theta))^2 + (b \sin(\theta))^2 = c^2$, expanding this:

$$a^2 - 2ab + b^2 \cos^2(\theta) + b^2 \sin^2(\theta) = c^2$$

$$a^2 - 2ab \cos(\theta) + b^2 = c^2$$

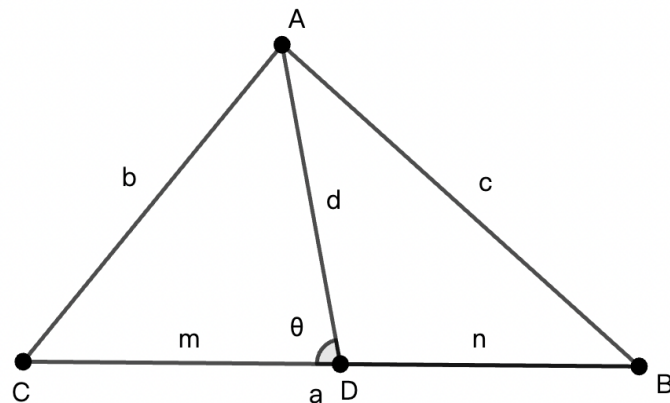
$$c^2 = a^2 + b^2 - 2ab \cos(\theta)$$

6 Stewart's Theorem

Stewart's Theorem, discovered by a Scottish mathematician named Matthew Stewart, shows a relationship between one cevian and three sides of a triangle.

Here is the theorem statement:

$$b^2n + c^2m = a(m + n) + d^2a$$



Note: a is the length of $\overline{CB}$.

6.1 Proof

Here is the setup:

$$AC = b, AB = c, CB = a, CD = m, DB = n, AD = d.$$

We will apply law of cosine twice on triangle ADC and triangle ADB to obtain the statement for Stewart's Theorem.

$$\text{On triangle ADC: } b^2 = d^2 + m^2 - 2dm \cos(\theta)$$

$$\text{On triangle ADB: } c^2 = d^2 + n^2 - 2dn \cos(\pi - \theta)$$

Multiply n on both sides of the first equation, multiply m on both sides of the second equation, we can obtain:

$$\begin{aligned} b^2n &= d^2n + m^2n - 2dmn \cos(\theta) \\ c^2m &= d^2m + n^2m - 2dmn \cos(\pi - \theta) \end{aligned}$$

As I mentioned earlier in this article, $\cos(\pi - \theta) = -\cos(\theta)$, we can do this substitution to the second equation:

$$c^2m = d^2m + n^2m + 2dmn \cos(\theta)$$

Now adding two equations together, we get:

$$b^2n + c^2m = d^2(m + n) + n^2m + m^2n$$

Since $m + n = a$ and we can factor out mn from $n^2m + m^2n$, we get the following:

$$b^2n + c^2m = d^2a + nma$$

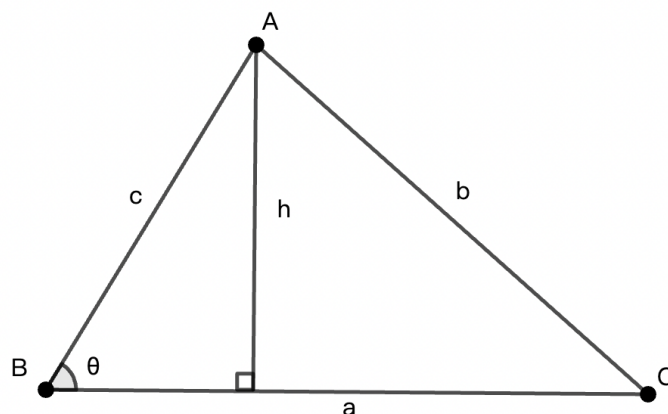
Therefore, the statement for Stewart's theorem is: $b^2n + c^2m = d^2a + amn$.

7 Alternative formula for triangle area using trigonometry

The most common area formula we used in math classes is probably $A = \frac{bh}{2}$, but there are other ways to find the area of a triangle!

Here is one of them that uses trigonometry:

$A = \frac{1}{2}ab \sin(\theta)$, where a and b are lengths of two sides and θ is the measure of the angle formed by these two sides.



The proof of this formula is straightforward. Indicated in the diagram, h is also $c \sin(\theta)$, then we can use the basic $\frac{bh}{2}$ formula. After substitution we can get: $\frac{ac \sin(\theta)}{2}$.

7.1 The case of equilateral triangle

You may heard the area formula for a equilateral triangle before, which is $\frac{s^2 \sqrt{3}}{4}$, where s is the side length. This comes from the formula derived above. Since all three angles of an equilateral triangle are 60 degrees, substitute this back to the formula, we get: $\frac{s \cdot s \cdot \sin(\frac{\pi}{3})}{2} = \frac{s^2 \sqrt{3}}{4}$.

8 Problems

1. Given triangle ABC with $AB = 3$, $AC = 4$, $m\angle BAC = 30^\circ$, find BC.
2. Find the area of a regular hexagon with side length 2.
3. Given triangle ABC with $AB = 6$, $AC = 9$, $BC = 10$, find the length of angle bisector AD, which bisects the angle BAC. (If you are able to derive a formula for the length of this segment, then you discovered the second angle bisector theorem)
4. Try to prove this: show that the centroid divides a median into two segments of a ratio of 1 : 2.
5. Derive a triangle area formula using perimeter and its inradius. (Hint: Look back to section 4.1)
6. In triangle ABC, $AB = 6$, $BC = 7$, and $CA = 8$. Point D lies on $\overline{BC}$, and $\overline{AD}$ bisects $\angle BAC$. Point E lies on $\overline{AC}$, and $\overline{BE}$ bisects $\angle ABC$. The bisectors intersect at F. What is the ratio AF : FD? (2016 AMC 12A Problem 12)
7. The lengths of the sides of a triangle are consecutive integers, and the largest angle is twice the smallest angle. What is the cosine of the smallest angle? (1982 AHSME Problem 23)

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